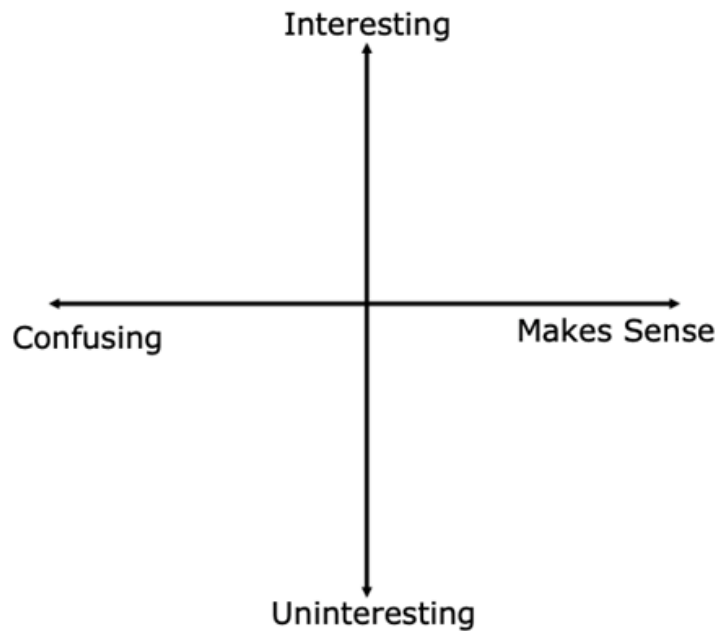




2.1.5 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest and understanding of today's learning targets.



2. Write one question you have about square values after today's lesson.

Unit 2, Lesson 2: Square Values and Square Roots



Benchmarks

MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.

MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.



Learning Targets

- ☐ I can determine the positive and negative solutions for $x^2 = p$, where p is a whole number.
- ☐ I can use the vocabulary related to radicals.
- ☐ I can interpret and evaluate a square root.



2.2.1 Warm-Up: Estimation

1. In 2006, scientists in North Wales found a deep-sea clam that was 507 years old. Deep-sea clams have the slowest growth rate, growing as little as 8.4 millimeters (mm), or one-third inch, every 100 years.



- a. Estimate the clam's size when the scientists found it in 2006.
- b. Explain your reasoning for your estimation.



2.2.2 Guided Instruction: Solving Equations in the Form $x^2 = p$

1. Consider the equation $x^2 = 1$.
 - a. Discuss with your partner what value(s) make the equation true.
 - b. Find the solutions for $x^2 = 1$.
 - c. Show the solutions that satisfy the equation.
2. Find the solutions for $64 = m^2$.
3. Find the solution for $x^2 = 0$.

For equations in the form $x^2 = p$, where p is a whole number, there are always two solutions, except in the case where $p = 0$. For all other whole number values of p , there is one positive and one negative solution.

4. Consider the equation $x^2 = -36$.

- a. Discuss with your partner whether you think it is possible to find a solution to this equation.
- b. Complete the statements.

There ☐ is
☐ is not a real number that can be

multiplied by itself to result in -36 . Therefore,

- ☐ there is not a real solution to the equation $x^2 = -36$.
- ☐ the real solution(s) to the equation $x^2 = -36$ is(are) _____.



2.2.3 Try It: Solving Equations in the Form

$$x^2 = p$$

Find the solution(s) for each equation.

1. $x^2 = 36$

2. $t^2 = 144$

3. $49 = n^2$



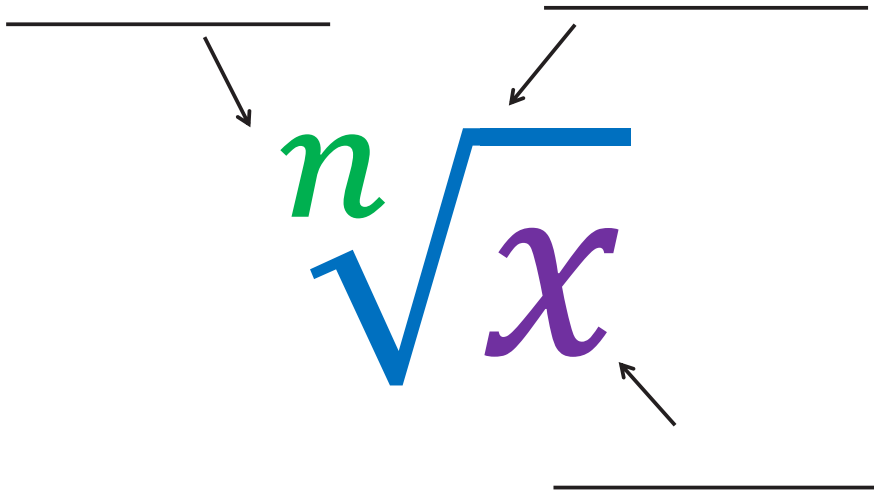
2.2.4 Guided Instruction: Square Roots

A perfect square is the product that results from squaring a number. Squaring a number is multiplying it by itself. The inverse operation to squaring a number is taking its **square root**.

The symbol $\sqrt{\quad}$ is called a **radical**. It is used to represent a **square root**. The value under the radical sign is called the **radicand**.

The **index** is the number outside the radical sign that indicates the root being taken.

1. Use this information to label the parts of a radical expression.



The symbol $\sqrt{\quad}$ is interpreted as the square root, unless an index other than 2 is indicated.



2. Consider the equations shown.

$$(4)^2 = 16$$

$$(-4)^2 = 16$$

The equations show which values can be squared to result in 16. Based on the information given about square roots, what is $\sqrt{16}$?

As explored in the previous lesson with the graph of $y = x^2$, every number has both a positive and negative square root, except for 0. The $\pm$ symbol is read, "plus or minus," and is used in front of the radical to denote both roots.

3. Complete each statement.

a. $\sqrt{4} = \pm 2$ because $\underline{\quad} \times \underline{\quad} = \underline{\quad}$ and $\underline{\quad} \times \underline{\quad} = \underline{\quad}$

b. $\sqrt{121} = \pm \underline{\quad}$ because $\underline{\quad} \times \underline{\quad} = \underline{\quad}$ and $\underline{\quad} \times \underline{\quad} = \underline{\quad}$

4. Explain why there is not a real number equivalent to $\sqrt{-81}$.



2.2.5 Guided Practice: Squares and Square Roots

1. Xavier completed two problems as shown.

Find the solution(s) for $b^2 = 25$.

$b = -5$ and $b = 5$

What is the value of $\sqrt{25}$? 5

- Discuss Xavier's answers with your partner.
- Complete the statements.

Xavier answered ☐ one problem
☐ both problems correctly.

There ☐ is only one solution
☐ are two solutions to the equation

$b^2 = 25$ because there ☐ is only one value
☐ are two values that

make(s) the equation true. When finding the value

of a square root, consider what value can be

multiplied by itself to result in the radicand.

Therefore, the value of $\sqrt{25}$ is ☐ 5.
☐ -5.
☐ ± 5 .

2. Find the value of each of the following.

a. $\sqrt{81} =$

b. $\sqrt{225} =$

c. $\sqrt{9} =$

d. $\sqrt{-64} =$

3. Solve each equation.

a. $a^2 = 169$

b. $225 = x^2$



2.2.6 Your Turn!

1. Find the value of each of the following.

a. $\sqrt{49} =$

b. $\sqrt{196} =$

2. Solve each equation.

a. $w^2 = 4$

b. $121 = k^2$



2.2.7 Wrap-Up: Reflection

1. Choose **two statements to complete** as you reflect on today's lesson.

a. One thing that made the most sense to me today was ...

b. One thing that I just don't understand is ...

c. One thing I learned today was ...

d. One thing I still wonder about is ...

e. One thing I noticed about square roots is ...